



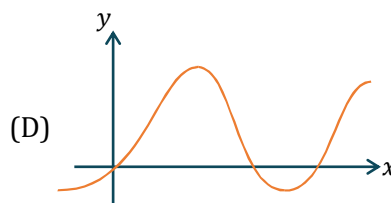
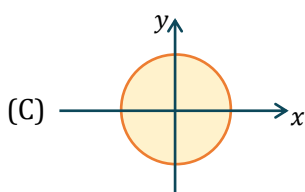
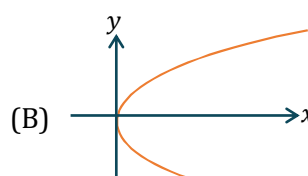
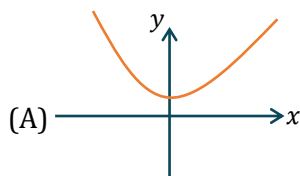
Functions : RACE-01

Single Correct Question

1. The range of the function  $f(x) = e^{-x} + e^x$ , is -  
 (A)  $f(x) \geq 1$  (B)  $f(x) \leq 1$  (C)  $f(x) \geq 2$  (D)  $f(x) \leq 2$  MFN148
2. The range of the function  $f: \mathbb{N} \rightarrow \mathbb{Z}; f(x) = (-1)^{x-1}$ , is -  
 (A)  $[-1, 1]$  (B)  $\{-1, 1\}$  (C)  $\{0, 1\}$  (D)  $\{0, 1, -1\}$  MFN149

Multiple Correct Question

3. Which of the following rule is/are function(s) ?  
 (A)  $f: \left(0, \frac{\pi}{2}\right) \rightarrow [0, 1], f(x) = \sin x$   
 (B)  $f: \{-3, -2, -1, 0, 1, 2, 3\} \rightarrow \{1, 4, 9\}, f(x) = x^2$   
 (C)  $f: \left[0, \frac{\pi}{2}\right) \rightarrow [0, 1) \rightarrow f(x) = \cos x$   
 (D)  $f: \left(0, \frac{\pi}{2}\right) \rightarrow (0, 1) \rightarrow f(x) = \cos x$  MFN150
4. Which of the following is (are) function ?  
 $A: \{1, 2, 3, 4\} \quad B = \{1, 4, 9, 16, 25\}$   
 (A)  $R_1 = \{(1, 1) (2, 4) (3, 9) (4, 16) (4, 25)\}$  (B)  $R_2 = \{(2, 4) (3, 9) (4, 16)\}$   
 (C)  $R_3 = \{(1, 1) (2, 4) (3, 9) (4, 16)\}$  (D)  $R_4 = \{(1, 1) (2, 4) (3, 9) (4, 4)\}$  MFN151
5. Which of the following represent graph(s) of a function ?



MFN152

Comprehension Type Question

Paragraph of Question No. 6 & 7

Graph of  $y = f(x)$  is drawn here, match Domain & Range of the  $f(x)$

| $y = f(x)$<br>Column-I |  | (Domain)<br>Column-II |                              | (Range)<br>Column-III |                              |
|------------------------|--|-----------------------|------------------------------|-----------------------|------------------------------|
| (P)                    |  | (I)                   | $(x_1, x_2] \cup (x_3, x_4]$ | (i)                   | $(y_1, y_2) \cup [y_3, y_4]$ |
|                        |  |                       |                              |                       |                              |
|                        |  |                       |                              |                       |                              |
|                        |  |                       |                              |                       |                              |
| (Q)                    |  | (II)                  | $(x_1, x_2) \cup [x_3, x_4]$ | (ii)                  | $(y_1, y_4]$                 |
| (R)                    |  | (III)                 | $[x_1, x_2) \cup [x_3, x_4)$ | (iii)                 | $[y_1, y_4)$                 |
| (S)                    |  | (IV)                  | $(x_1, x_2) \cup (x_3, x_4]$ | (iv)                  | $(y_1, y_4) - \{y_2\}$       |

6. Select the correct match

(A) P-(II)-(ii)

(B) Q-(I)-(ii)

(C) R-(III)-(ii)

(D) S-(II)-(iv)

MFN153

7. Select the correct match

(A) P-(II)-(i)

(B) Q-(II)-(i)

(C) R-(III)-(ii)

(D) S-(I)-(ii)

MFN154

Matrix Match Type Question

8. Match the function given in **column-I** with corresponding domain in **column-II**.

Column-I

Column-II

(P)  $y = \sqrt{\frac{x(x-1)^3(x-5)}{(x+4)^2(x-2)}}$

(1)  $x \in [-2, 0] \cup [2, \infty)$

(Q)  $y = \sqrt{(e^x - 1)(x^2 - 4)}$

(2)  $-1 \leq x \leq 1$

(R)  $y = \sqrt{-px} \quad (p > 0)$

(3)  $x \in (-\infty, -4) \cup (-4, 0] \cup [1, 2) \cup [5, \infty)$

(S)  $y = \sqrt{1 - |x|}$

(4)  $-\infty < x \leq 0$

MFN155

9. Match the function given in **column-I** with corresponding domain in **column-II**.

Column-I

Column-II

(P)  $f(x) = \sqrt{|x| - x}$

(1)  $x \in \left(-\infty, -\frac{1}{2}\right] \cup (0, 1) \cup (2, \infty)$

(Q)  $f(x) = \frac{1}{\sqrt{|x| - x}}$

(2)  $x \in (-\infty, 0) \cup (4, \infty)$

(R)  $f(x) = \sqrt{\frac{(2x+1)}{x^3 - 3x^2 + 2x}}$

(3)  $x \in (-\infty, 0)$

(S)  $f(x) = \frac{1}{\sqrt{x^2 - 4x}}$

(4)  $x \in \mathbb{R}$

MFN156

10. Match the function given in **column-I** with corresponding domain in **column-II**.

Column-I

Column-II

(P)  $y = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$

(1)  $2k\pi < x < (2k+1)\pi$ , where  $k$  is an integer

(Q)  $y = \frac{1}{\sqrt{\sin x}} + \sqrt[3]{\sin x}$

(2)  $4 \leq x \leq 6$

(R)  $y = \log_{10}(\sqrt{x-4} + \sqrt{6-x})$

(3)  $2 < x < 3$

(S)  $y = \log_{10}[1 - \log_{10}(x^2 - 5x + 16)]$

(4)  $-1 < x < 0$  or  $1 < x < 2$  or  $2 < x < \infty$

MFN157

11. Match the function given in **column-I** with corresponding domain in **column-II**.

Column-I

Column-II

(P)  $y = \log_x 2$

(1)  $-2 \leq x < 0$  or  $0 < x < 1$

(Q)  $y = \frac{1}{\log_{10}(1-x)} + \sqrt{x+2}$

(2)  $\frac{3}{2} < x < 2$  or  $2 < x < \infty$

(R)  $y = \sqrt{x} + \sqrt[3]{\frac{1}{x-2}} - \log_{10}(2x-3)$

(3)  $x \in \mathbb{R}$

(S)  $y = \frac{1}{x^2 + 1}$

(4)  $0 < x < 1$  or  $1 < x < \infty$

MFN158

Integer Type Question

12. If  $f(x) = 3x + 4$ ;  $g(x) = \begin{cases} x^2 + 1; & x \geq 1 \\ 2x + 3; & x < 1 \end{cases}$  and  $h(x) = \begin{cases} e^x & ; \quad x \geq 5 \\ 2x^2 + 3x & ; \quad 2 < x < 5 \\ 5 - x & ; \quad x < 2 \end{cases}$  then find  $h(f(g(-2)))$

MFN159

ANSWER KEY

| Question | 1                                                                    | 2 | 3   | 4                                                                    | 5   | 6 | 7                                                                    | 8                                                                    |    |
|----------|----------------------------------------------------------------------|---|-----|----------------------------------------------------------------------|-----|---|----------------------------------------------------------------------|----------------------------------------------------------------------|----|
| Answer   | C                                                                    | B | A,D | C,D                                                                  | A,D | B | A                                                                    | $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$ |    |
| Question | 9                                                                    |   |     | 10                                                                   |     |   | 11                                                                   |                                                                      | 12 |
| Answer   | $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$ |   |     | $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$ |     |   | $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$ |                                                                      | 4  |



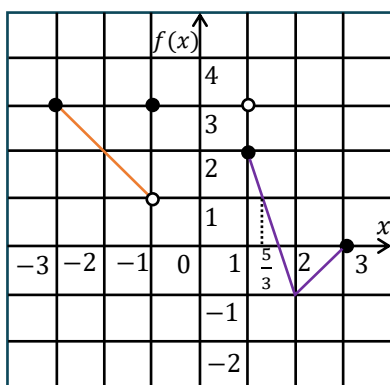
Functions : RACE-02

Single Correct Question

1. Arrange the following in decreasing order  $3^x, 5^x, 2^x, 4^x$  if  $x > 0$   
 (A)  $3^x > 5^x > 2^x > 4^x$  (B)  $5^x > 4^x > 3^x > 2^x$   
 (C)  $2^x > 3^x > 4^x > 5^x$  (D)  $2^x > 5^x > 4^x > 3^x$   
**MFN160**
2. Find the domain of the function  $f(x) = \frac{1}{\sqrt{[\![x]\!]-7}} - 11$ , where  $[\cdot]$  denotes greatest integer function.  
 (A)  $x \in (-\infty, -17] \cup [19, \infty)$  (B)  $x \in (-\infty, -19] \cup [17, \infty)$   
 (C)  $x \in (-\infty, -19] \cup [19, \infty)$  (D)  $x \in (-\infty, -20] \cup [19, \infty)$   
**MFN161**
3. If  $y = 2[x] + 1 = 3[x + 1] - 5$  then find  $[x + 2y]$ , where  $[\cdot]$  denotes greatest integer function.  
 (A) 17 (B) 18 (C) 19 (D) 20  
**MFN162**
4. Find the range of  $x$  if  $[x]^2 - [x] - 2 = 0$ , where  $[\cdot]$  denotes greatest integer function.  
 (A)  $x \in [1, 0) \cup [2, 3)$  (B)  $x \in [-1, 0) \cup [2, 3)$   
 (C)  $x \in [-1, 0) \cup [-2, 3)$  (D)  $x \in [-1, 0) \cup [2, 3)$   
**MFN163**

Comprehension Type Question

Solve the following inequalities using graph of  $f(x)$ :



5. Find the values of  $x$  for  $0 \leq f(x) \leq 1$   
 (A)  $\left[\frac{4}{3}, \frac{5}{3}\right] \cup \{3\}$  (B)  $[-2, -1) \cup [1, 3]$  (C)  $[-3, -2] \cup [-1, 1]$  (D)  $\left(\frac{5}{3}, 3\right) - \{2\}$   
**MFN164**
6. Find the values of  $x$  for  $-1 \leq f(x) \leq 2$   
 (A)  $\left[\frac{4}{3}, \frac{5}{3}\right] \cup \{3\}$  (B)  $[-2, -1) \cup [1, 3]$  (C)  $[-3, -2] \cup [-1, 1]$  (D)  $\left(\frac{5}{3}, 3\right) - \{2\}$   
**MFN165**

7. Find the values of  $x$  for  $f(x) > -1$  &  $f(x) < 0$

- (A)  $\left[\frac{4}{3}, \frac{5}{3}\right] \cup \{3\}$  (B)  $[-2, -1) \cup [1, 3]$  (C)  $[-3, -2] \cup [-1, 1]$  (D)  $\left(\frac{5}{3}, 3\right) - \{2\}$

MFN166

**Matrix Match Type Question**

8. **Column-I**

**Column-II**

(P) Find range of the following functions  
 $f: R \rightarrow R, f(x) = x^2 - 8x + 7$

(1)  $[-8, 16]$

(Q) Find range of the following functions  
 $f: [-1, 3] \rightarrow R, f(x) = x^2 - 8x + 7$

(2)  $[-8, \infty)$

(R) Find range of the following functions  
 $f: [-1, \infty) \rightarrow R, f(x) = x^2 - 8x + 7$

(3)  $[-9, \infty)$

(S) Find range of the following functions  
 $f: (-\infty, 3] \rightarrow R, f(x) = x^2 - 8x + 7$

(4)  $[16, \infty)$

MFN167

9. Find the range of the following functions :

**Column-I**

**Column-II**

(P)  $f(x) = 2^{x^2} + 1$

(1)  $\left[\frac{1}{11}, \frac{1}{5}\right]$

(Q)  $f(x) = \frac{1}{8 - 3\sin x}$

(2)  $\left[\frac{1}{3}, 1\right]$

(R)  $f(x) = \frac{1}{2 - \cos 3x}$

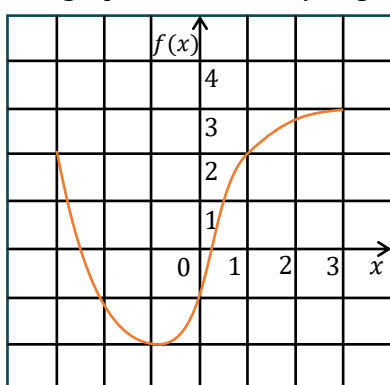
(3)  $(0, 1]$

(S)  $f(x) = \frac{1}{x^2 + 1}$

(4)  $[2, \infty)$

MFN168

10. The graph of a function  $f$  is given.



**Column-I**

**Column-II**

(P) State the value of  $f(-1)$ .

(1)  $-3, 1$

(Q) For what values of  $x$  is  $f(x) = 2$

(2)  $[-1, 3]$

(R) State the domain and range of  $f$ .

(3)  $[-3, 3], [-2, 3]$

(S) On what interval is  $f$  increasing ?

(4)  $-2$

MFN169

**Subjective Type Question**

11. Draw the graph of following

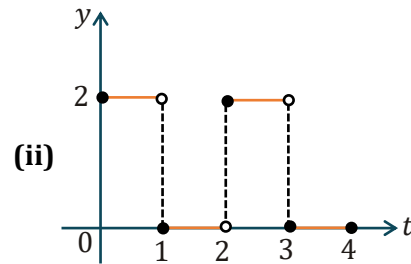
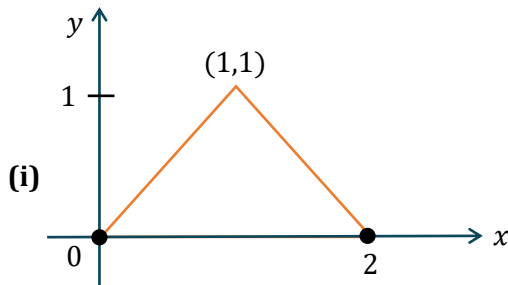
(i)  $y = |x + 3| + |x - 1| + |x - 5|$

(ii)  $f(x) = [e^x]$ , where  $[.]$  denotes greatest integer function.

(iii)  $f(x) = \begin{cases} x+1; & -1 \leq x < 4 \\ -x; & 4 \leq x < 7 \end{cases}$

MFN170

12. Find a formula for each function graphed



MFN171

**ANSWER KEY**

| Question | 1                  | 2 | 3 | 4 | 5 | 6                  | 7 | 8                  |
|----------|--------------------|---|---|---|---|--------------------|---|--------------------|
| Answer   | B                  | C | A | B | A | B                  | D | P→3, Q→1, R→3, S→2 |
| Question | 9                  |   |   |   |   | 10                 |   |                    |
| Answer   | P→4, Q→1, R→2, S→3 |   |   |   |   | P→4, Q→1, R→3, S→2 |   |                    |



Functions : RACE-03

Single Correct Question

1. If  $[x]$  and  $\{x\}$  denotes the greatest integer function less than or equal to  $x$  and fractional part function respectively, then the number of real  $x$ , satisfying the equation  $(x - 2)[x] = \{x\} - 1$ , is-
- (A) 0 (B) 1 (C) 2 (D) infinite

MFN172

2. If  $[a]$  denotes the greatest integer less than or equal to  $a$  and  $-1 \leq x < 0, 0 \leq y < 1, 1 \leq z < 2$ ,

then  $\begin{vmatrix} [x]+1 & [y] & [z] \\ [x] & [y]+1 & [z] \\ [x] & [y] & [z]+1 \end{vmatrix}$  is equal to -

- (A)  $[x]$  (B)  $[y]$  (C)  $[z]$  (D) none of these

MFN173

3. The domain of definition of  $f(x) = \frac{\log_2(x+3)}{x^2+3x+2}$  is :

- (A)  $R \setminus \{-1, -2\}$  (B)  $(-2, \infty)$  (C)  $R \setminus \{-1, -2, -3\}$  (D)  $(-3, \infty) \setminus \{-1, -2\}$

MFN174

4. The number of integers lying in the domain of the function  $f(x) = \sqrt{\log_{0.5}\left(\frac{5-2x}{x}\right)}$  is -

- (A) 3 (B) 2 (C) 1 (D) 0

MFN175

Comprehension Type Question

Paragraph (Q. 5 to 7)

Let  $\{x\}$  &  $[x]$  denote the fractional and integral part of a real number  $x$  respectively.

5. Find  $x$  if  $2x + \{x\} = 5[x]$

- (A)  $x \in \{0\}$  (B)  $x \in \{0,1\}$  (C)  $x \in \{1,2\}$  (D)  $x \in \{0,2\}$

MFN176

6. Find  $x$  if  $x - \{x\} = 2[x]$

- (A)  $x \in [-1, 1)$  (B)  $x \in [0, 2)$  (C)  $x \in [0, 1)$  (D)  $x \in [0, 1]$

MFN177

7. Find the number of values of  $x$  if  $4\{x\} = x + [x]$ .

- (A) 1 (B) 2 (C) 3 (D) 4

MFN178



Matrix Match Type Question

8. Let  $\{x\}$  &  $[x]$  denotes the fraction and integral part of a real number  $x$  respectively, then match the column.

Column-I

- (P)  $[x^2] \geq 4$   
 (Q)  $[x]^2 - 5[x] + 6 = 0$   
 (R)  $x = \{x\}$   
 (S)  $[x] < -5$

Column-II

- (1)  $x \in [2, 4)$   
 (2)  $x \in (-\infty, -2] \cup [2, \infty)$   
 (3)  $x \in (-\infty, -5)$   
 (4)  $x \in \{-2\}$   
 (5)  $x \in [0, 1)$

MFN179

9. Find the range of the following functions :

Column-I

- (P)  $f(x) = e^{(x-1)^2}$   
 (Q)  $f(x) = x^3 - x^2 + x + 1$   
 (R)  $f(x) = \log(x^8 + x^4 + x^2 + 1)$   
 (S)  $f(x) = \frac{7x+9}{3x-5}$

Column-II

- (1)  $[0, \infty)$   
 (2)  $R - \left\{\frac{7}{3}\right\}$   
 (3)  $\mathbb{R}$   
 (4)  $[1, \infty)$

MFN180

10. Find the domains of definitions of the following functions :

Column-I

- (P)  $f(x) = \log_{3-2\{x\}}(x^2 - 5x + 13)$   
 (Q)  $f(x) = \sqrt{\frac{9-3^x}{5^{-x}-125}}$   
 (R)  $f(x) = \sqrt{x^2 - |x|} + \frac{1}{\sqrt{9-x^2}}$   
 (S)  $f(x) = \sqrt{(x^2 - 3x - 10) \ln^2(x-3)}$

Column-II

- (1)  $x \in (-\infty, -3) \cup [2, \infty)$   
 (2)  $\{4\} \cup [5, \infty)$   
 (3)  $x \in R$   
 (4)  $[-3, -1] \cup \{0\} \cup [1, 3)$

MFN181

11. Match the graph of function given in **Column-I** with sketch in **Column-II** :

**Column-I**

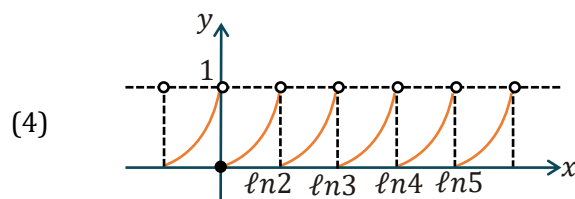
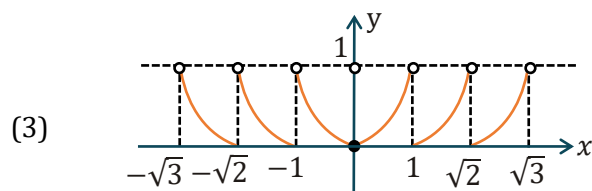
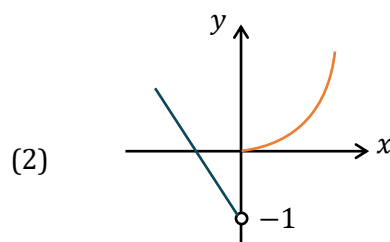
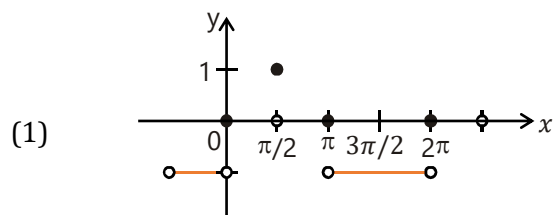
(P)  $f(x) = \begin{cases} x^2; & x \geq 0 \\ -x-1; & x < 0 \end{cases}$

(Q)  $f(x) = [\sin x]$

(R)  $f(x) = \{e^x\}$

(S)  $f(x) = \{x^2\}$

**Column-II**



MFN182

### Integer Type Question

12. Find the number of integer value of  $x$  in domain of function :

$$f(x) = \sin(\sqrt{1-x^2}) + \sqrt{x+2} + \frac{1}{\log_{10}(x+1)}$$

MFN183

### ANSWER KEY

| Question | 1                                                                    | 2 | 3 | 4 | 5                                                                    | 6 | 7 | 8                                                                                   |    |
|----------|----------------------------------------------------------------------|---|---|---|----------------------------------------------------------------------|---|---|-------------------------------------------------------------------------------------|----|
| Answer   | D                                                                    | C | D | C | A                                                                    | C | B | $P \rightarrow 1, 2, 3, 4; Q \rightarrow 1, 2; R \rightarrow 5; S \rightarrow 2, 3$ |    |
| Question | 9                                                                    |   |   |   | 10                                                                   |   |   | 11                                                                                  | 12 |
| Answer   | $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$ |   |   |   | $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$ |   |   | $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$                | 1  |



Functions : RACE-04

Single Correct Question

1. A function  $f$  has domain  $[-1, 2]$  and range  $[0, 1]$ . The domain and range respectively of the function  $g$  defined by  $g(x) = 1 - f(x + 1)$  is  
 (A)  $[-1, 1]; [-1, 0]$  (B)  $[-2, 1]; [0, 1]$  (C)  $[0, 2]; [-1, 0]$  (D)  $[1, 3]; [-1, 0]$

MFN184

2. The range of the function  $f(x) = \text{sgn}\left(\frac{\sin^2 x + 2\sin x + 4}{\sin^2 x + 2\sin x + 3}\right)$  is (where  $\text{sgn}(\cdot)$  denotes signum function)-  
 (A)  $\{-1, 0, 1\}$  (B)  $\{-1, 0\}$  (C)  $\{1\}$  (D)  $\{0, 1\}$

MFN185

3. The range of the function  $f(x) = \sqrt{4-x^2} + \sqrt{x^2-1}$  is  
 (A)  $[\sqrt{3}, \sqrt{7}]$  (B)  $[\sqrt{3}, \sqrt{5}]$  (C)  $[\sqrt{2}, \sqrt{3}]$  (D)  $[\sqrt{3}, \sqrt{6}]$

MFN186

4. Range of the function  $f(x) = \frac{x^2 + x + 2}{x^2 + x + 1}$  is -  
 (A)  $[1, 2]$  (B)  $[1, \infty)$  (C)  $\left[2, \frac{7}{3}\right]$  (D)  $\left(1, \frac{7}{3}\right]$

MFN187

5. Range of the function  $f(x) = \frac{\{x\}}{1 + \{x\}}$ , where  $\{x\}$  denotes the fractional part function, is  
 (A)  $[0, 1)$  (B)  $\left[0, \frac{1}{2}\right]$  (C)  $\left[0, \frac{1}{2}\right)$  (D)  $\left(0, \frac{1}{2}\right]$

MFN188

6. Which of the following function is implicit function ?  
 (A)  $xy - \cos(x + y) = 0$  (B)  $y = x^3$   
 (C)  $y = \log(x^2 + x + 1)$  (D)  $y = |x|$

MFN189

7. For the function  $f(x) = \frac{e^x + 1}{e^x - 1}$ , if  $n(d)$  denotes the number of integers which are not in its domain and  $n(r)$  denotes the number of integers which are not in its range, then  $n(d) + n(r)$  is equal to -  
 (A) 2 (B) 3 (C) 4 (D) Infinite

MFN190

8. Find range of  $f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$   
 (A)  $(-1, 1)$  (B)  $[-1, 1]$  (C)  $\{-1, 1\}$  (D)  $[-1, 1]$

MFN191

9. Find range of  $f(x) = \frac{(\ln x)^2 - 5\ln x + 4}{(\ln x)^2 - 2\ln x - 3}$

(A)  $\mathbb{R}^+$

(B)  $\mathbb{R}^-$

(C)  $\mathbb{R}$

(D)  $\mathbb{R}_0$

MFN192

**Multiple Correct Question**

10. The range of the function  $f(\theta) = \sqrt{8\sin^2 \theta + 4\cos^2 \theta - 8\sin \theta \cos \theta}$  is -

(A)  $[\sqrt{5}-1, \sqrt{5}+1]$

(B)  $[0, \sqrt{5}+1]$

(C)  $[\sqrt{6}-\sqrt{20}, \sqrt{6}+\sqrt{20}]$

(D) none of these

MFN193

**Matrix Match Type Question**

11. Find the range of the following function :

**Column-I**

(P)  $\log_4 \left| x + \frac{1}{x} \right|$

(Q)  $f(x) = \sin(3x^2 + 1)$

(R)  $f(x) = \frac{x^2 - 5x + 6}{4x^2 - 5x - 6}$

(S)  $f(x) = \frac{3x-6}{x^3 - 4x^2 + 2x + 4}$

**Column-II**

(1)  $(-\infty - 1] \cup (0, \infty) - \left\{ \frac{-3}{2} \right\}$

(2)  $R - \left\{ \frac{1}{4}, \frac{-1}{11} \right\}$

(3)  $\left[ \frac{1}{2}, \infty \right)$

(4)  $[-1, 1]$

MFN194

**Integer Type Question**

12. Let  $f(x) = (x+1)(x+2)(x+3)(x+4) + 5$  where  $x \in [-6, 6]$ . If the range of the function is  $[a, b]$  where  $a, b \in \mathbb{N}$  then find the value of  $(a+b)$ .

MFN195

**ANSWER KEY**

| Question | 1                  | 2 | 3 | 4 | 5    | 6 | 7 | 8 | 9 | 10  |
|----------|--------------------|---|---|---|------|---|---|---|---|-----|
| Answer   | B                  | C | D | D | C    | A | C | A | C | A,C |
| Question | 11                 |   |   |   | 12   |   |   |   |   |     |
| Answer   | P→3; Q→4; R→2; S→1 |   |   |   | 5049 |   |   |   |   |     |



Functions : RACE-05

Single Correct Question

1. If  $2f(x) - 3f\left(\frac{1}{x}\right) = x^2$ ,  $x$  is not equal to zero, then  $f(2)$  is equal to-

(A)  $-\frac{7}{4}$  (B)  $\frac{5}{2}$  (C)  $-1$  (D) none of these

MFN196

2. If  $f(x) = \frac{4^x}{4^x + 2}$ , then  $f(x) + f(1-x)$  is equal to-

(A) 0 (B)  $-1$  (C) 1 (D) 4

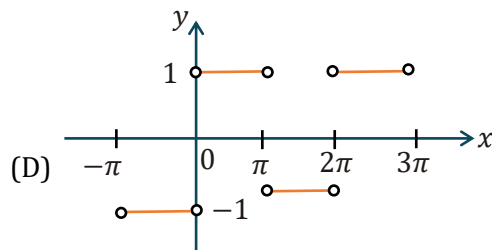
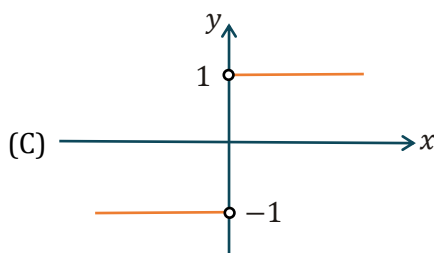
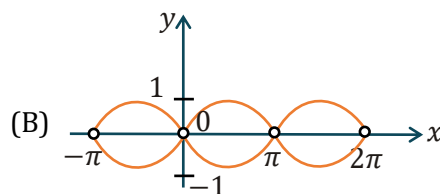
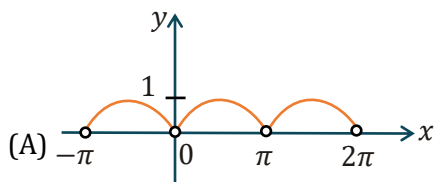
MFN197

3. If  $x^4 f(x) - \sqrt{1 - \sin 2\pi x} = |f(x)| - 2f(x)$ , then  $f(-2)$  equals

(A)  $\frac{1}{17}$  (B)  $\frac{1}{11}$  (C)  $\frac{1}{19}$  (D) 0

MFN198

4. Which of the following is the graph of  $y = \frac{|\sin x|}{\sin x}$ ?



MFN199

5. A lion moves in the region given by the graph  $y - |y| - x + |x| = 0$ . Then on which of the following curve a person can move so that he does not encounter lion -

(A)  $y = e^{-|x|}$  (B)  $y = \frac{1}{x}$  (C)  $y = \text{signum}(x)$  (D)  $y = -|4 + |x||$

MFN200

Functions

6. Let function  $f: R \rightarrow R$  be defined by  $f(x) = 2x + \sin x$  for  $x \in R$ . Then  $f$  is -  
 (A) one to one and onto  
 (B) one to one but not onto  
 (C) onto but not one to one  
 (D) neither one to one nor onto

MFN201

7. Let  $f(x) = \frac{x}{1+x}$  defined from  $(0, \infty) \rightarrow [0, \infty)$  then by  $f(x)$  is -  
 (A) one-one but not onto  
 (B) one-one and onto  
 (C) many one but not onto  
 (D) many one and onto

MFN202

8. Let  $f: N \rightarrow N$  be defined as  $f(n) = \begin{cases} \frac{n+1}{2}, & \text{if } n \text{ is odd} \\ \frac{n}{2}, & \text{if } n \text{ is even} \end{cases}$  for all  $n \in N$ . Then  $f$  is -  
 (A) injective but not surjective  
 (B) surjective but not injective  
 (C) both injective as well as surjective  
 (D) neither injective nor surjective

MFN203

9. Let  $E = \{1, 2, 3, 4\}$  &  $F = \{1, 2\}$ . Then the number of onto functions from  $E$  to  $F$  is -  
 (A) 14 (B) 16 (C) 12 (D) 8

MFN204

Multiple Correct Question

10. If a function is defined by an implicit equation  $2^{|x|+|y|} + 2^{|x|-|y|} = 2$ , then -  
 (A) Domain of function is singleton  
 (B) Range of function is singleton  
 (C) graph of the function intersects the line  $y = x$   
 (D) maximum value of function is 2

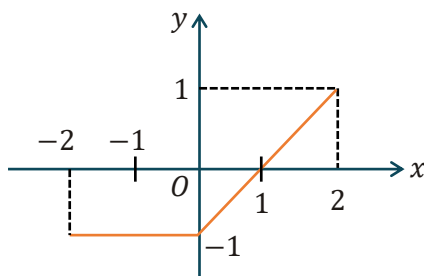
MFN205

11. Which of the following pair of functions are identical?  
 (A)  $f(x) = \operatorname{sgn}(x^2 - 3x + 4)$  and  $g(x) = e^{\{x\}}$   
 (B)  $f(x) = \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}}$  and  $g(x) = \tan x$   
 (C)  $f(x) = \ln(1+x) + \ln(1-x)$  and  $g(x) = \ln(1-x^2)$   
 (D)  $f(x) = \frac{\cos x}{1 - \sin x}$  and  $g(x) = \frac{1 + \sin x}{\cos x}$

MFN206

Matrix Match Type Question

12. The graph of the function  $y = f(x)$  is as follows :



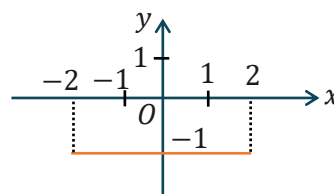
Match the function mentioned in **Column-I** with the respective graph given in **Column-II**.

Column-I

Column-II

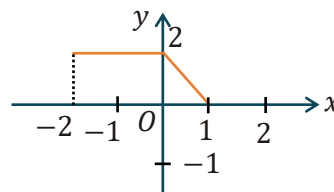
(P)  $y = |f(x)|$

(1)



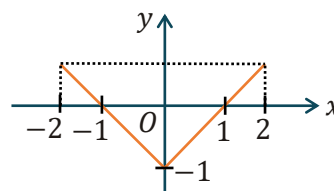
(Q)  $y = f(|x|)$

(2)



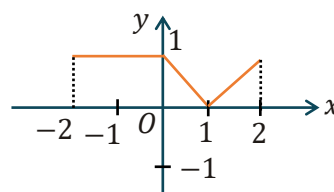
(R)  $y = f(-|x|)$

(3)



(S)  $y = (|f(x)| - f(x))$

(4)



MFN207

ANSWER KEY

| Question | 1   | 2                  | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10    |
|----------|-----|--------------------|---|---|---|---|---|---|---|-------|
| Answer   | A   | C                  | A | D | D | A | A | B | A | A,B,C |
| Question | 11  | 12                 |   |   |   |   |   |   |   |       |
| Answer   | A,C | P→4; Q→3; R→1; S→2 |   |   |   |   |   |   |   |       |



Functions : RACE-06

Single Correct Question

1. Which of the following function is surjective but not injective  
 (A)  $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^4 + 2x^3 - x^2 + 1$       (B)  $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^3 + x + 1$   
 (C)  $f: \mathbb{R} \rightarrow \mathbb{R}^+ \quad f(x) = \sqrt{1+x^2}$       (D)  $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^3 + 2x^2 - x + 1$   
**MFN208**
2. A mapping  $f: A \rightarrow [-1, 1]$  defined by  $f(x) = \sin x, \forall x \in \mathbb{R}$ , where  $A$  is a subset of  $\mathbb{R}$  (the set of all real numbers) is one-one and onto if  $A$  is the interval, then  $A$  belongs to  
 (A)  $[0, 2\pi]$       (B)  $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$       (C)  $[-\pi, \pi]$       (D)  $[0, \pi]$   
**MFN209**
3. Let  $f(x) = \frac{\alpha x}{x+1}, \forall x \in D_f$ , Then for what value of  $\alpha$  is  $f(f(x)) = x$ ?  
 (A)  $\sqrt{2}$       (B)  $-\sqrt{2}$       (C) 1      (D) -1  
**MFN210**
4. Let  $g(x) = 1 + x - [x]$  &  $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$ . Then for all  $x, f(g(x))$  is equal to,  
 where  $[.]$  denotes greatest integer function.  
 (A)  $x$       (B) 1      (C)  $f(x)$       (D)  $g(x)$   
**MFN211**
5.  $f(x) = x^3 - x$  &  $g(x) = \sin 2x$ , then select the incorrect option-  
 (A)  $f(f(1)) = 0$       (B)  $f(f(-1)) = 0$       (C)  $f\left(g\left(\frac{\pi}{2}\right)\right) = 0$       (D)  $f\left(g\left(\frac{\pi}{4}\right)\right) = 1$   
**MFN212**
6. If  $f$  be the greatest integer function and  $g$  be the modulus function, then  
 $(g \circ f)\left(-\frac{5}{3}\right) - (f \circ g)\left(-\frac{5}{3}\right) =$   
 (A) 1      (B) -1      (C) 2      (D) 4  
**MFN213**
7. A function  $f: \mathbb{R} \rightarrow \mathbb{R}$  is such that  $f\left(\frac{1-x}{1+x}\right) = x$  for all  $x \neq -1$ . Then which of the following not correct.  
 (A)  $f(f(x)) = x$       (B)  $f\left(\frac{1}{x}\right) = -f(x), x \neq 0$   
 (C)  $f(-x-2) = -f(x) - 2$       (D)  $f(-x-2) = f(x) - 2$ .  
**MFN214**



**Multiple Correct Question**

8.  $f(x) = \begin{cases} [x], & -2 \leq x \leq -1 \\ |x|+1, & -1 < x \leq 2 \end{cases}$  and  $g(x) = \begin{cases} [x], & -\pi \leq x \leq 0 \\ \sin x, & 0 \leq x \leq \pi \end{cases}$ ,  $[.]$  denotes the greatest integer function. If  $h(x) = g(f(x))$
- (A) domain of  $h(x)$  is  $[-1, 1]$   
 (B) Range of  $h(x)$  is  $\{-2, -1\} \cup [\sin 3, 1]$   
 (C) Domain of  $h(x)$  is  $[-2, 2]$   
 (D) Range of  $h(x)$  is  $\{-2, -1\} \cup [\sin 2, 1]$

MFN215

**Matrix Match Type Question**

9. Match the functions given in column-I correctly with mappings given in column-II.

**Column-I**

**Column-II**

(P)  $f: \left[-\frac{1}{2}, \frac{1}{2}\right] \rightarrow \left[\frac{4}{7}, \frac{4}{3}\right]$

(1) Injective mapping

$f(x) = \frac{1}{x^2 + x + 1}$

(Q)  $f: [-2, 2] \rightarrow [-1, 1]$

(2) Non-injective mapping

$f(x) = \sin x$

(R)  $f: \mathbb{R} - I \rightarrow \mathbb{R}$   $f(x) = \ell n\{x\}$ ,

(3) Surjective mapping

(where  $\{.\}$  represents fractional part function)

(S)  $f: (-\infty, 0] \rightarrow [1, \infty)$ ,  $f(x) = (1 + \sqrt{-x}) + (\sqrt{-x} - x)$

(4) Non-surjective mapping

(5) Bijective mapping

MFN216

**Integer Type Question**

10. If  $f(x) = \begin{cases} 0 & x < 1 \\ 2x-2 & x \geq 1 \end{cases}$ ; then the number of solutions of the equation  $f(f(f(x))) = x$  is

MFN217

**ANSWER KEY**

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8   | 9                                 | 10 |
|----------|---|---|---|---|---|---|---|-----|-----------------------------------|----|
| Answer   | D | B | D | B | D | A | D | B,C | P→1,3,5; Q→2,3;<br>R→2,4; S→1,3,5 | 2  |



Functions : RACE-07

Single Correct Question

1. If  $f: (-\infty, 3] \rightarrow [7, \infty)$ ;  $f(x) = x^2 - 6x + 16$ , then which of the following is true -

(A)  $f^{-1}(x) = 3 + \sqrt{x - 7}$

(B)  $f^{-1}(x) = 3 - \sqrt{x - 7}$

(C)  $f^{-1}(x) = \frac{1}{x^2 - 6x + 16}$

(D)  $f$  is many-one

MFN218

2. If  $f: [1, \infty) \rightarrow [2, \infty)$  is given by,  $f(x) = x + \frac{1}{x}$ , then  $f^{-1}(x)$  equals -

(A)  $\frac{x + \sqrt{x^2 - 4}}{2}$

(B)  $\frac{x}{1 + x^2}$

(C)  $\frac{x - \sqrt{x^2 - 4}}{2}$

(D)  $1 - \sqrt{x^2 - 4}$

MFN219

3. Suppose  $f(x) = (x + 1)^2$  for  $x \geq -1$ . If  $g(x)$  is the function whose graph is the reflection of the graph of  $f(x)$  with respect to the line  $y = x$ , then  $g(x)$  equals -

(A)  $-\sqrt{x} - 1, x \geq 0$  (B)  $\frac{1}{(1+x)^2}, x \geq -1$  (C)  $\sqrt{x+1}, x \geq -1$  (D)  $\sqrt{x} - 1, x \geq 0$

MFN220

4.  $f: \mathbb{R} \rightarrow \mathbb{R}$  such that  $f(x) = \ln(x + \sqrt{x^2 + 1})$ . Another function  $g(x)$  is defined such that  $g \circ f(x) = x, \forall x \in \mathbb{R}$ . Then  $g(2)$  is -

(A)  $\frac{e^2 + e^{-2}}{2}$

(B)  $e^2$

(C)  $\frac{e^2 - e^{-2}}{2}$

(D)  $e^{-2}$

MFN221

5. If  $f(x) = x|x|$  then  $f^{-1}(x)$  equals-

(A)  $\sqrt{|x|}$

(B)  $(\operatorname{sgn} x) \cdot \sqrt{|x|}$

(C)  $-\sqrt{|x|}$

(D) Does not exist

(where  $\operatorname{sgn}(x)$  denotes signum function of  $x$ )

MFN222

6. Let  $f$  be a function defined by  $f(x) = (x - 1)^2 + 1, (x \geq 1)$

**Statement - 1 :** The set  $\{x: f(x) = f^{-1}(x)\} = \{1, 2\}$

**Statement - 2 :**  $f$  is bijection and  $f^{-1}(x) = 1 + \sqrt{x - 1}, x \geq 1$ .

(A) Statement-1 is true, Statement-2 is false.

(B) Statement-1 is false, Statement-2 is true.

(C) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1.

(D) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for statement-1.

MFN223

7. If  $f(x) = 3x - 5$ , then  $f^{-1}(x)$  -  
 (A) is given by  $\frac{1}{3x-5}$  (B) is given by  $\frac{x+5}{3}$   
 (C) does not exist because  $f$  is not one-one (D) does not exist because  $f$  is not onto

MFN224

8. If the function  $f: [1, \infty) \rightarrow [1, \infty)$  is defined by  $f(x) = 2^{x(x-1)}$ , then  $f^{-1}(x)$  is -  
 (A)  $\left(\frac{1}{2}\right)^{x(x-1)}$  (B)  $\frac{1}{2}\left(1 + \sqrt{1 + 4\log_2 x}\right)$   
 (C)  $\frac{1}{2}\left(1 - \sqrt{1 + 4\log_2 x}\right)$  (D) Not defined

MFN225

### Matrix Match Type Question

- | 9.  | Column-I                                                   | Column-II                         |
|-----|------------------------------------------------------------|-----------------------------------|
| (P) | $f(x) = \frac{x(a^x + 1)}{a^x - 1}$                        | (1) Neither even nor odd function |
| (Q) | $f(x) = x\sin^2 x - x^3$                                   | (2) Even function                 |
| (R) | $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$               | (3) Odd function                  |
| (S) | $f(x) = [(x+1)^2]^{\frac{1}{3}} + [(x-1)^2]^{\frac{1}{3}}$ | (4) Identity function             |

MFN226

### Integer Correct Type Question

10. Let  $a$  and  $b$  be real numbers and let  $f(x) = a\sin x + b\sqrt[3]{x} + 4, \forall x \in \mathbb{R}$ .  
 If  $f(\log_{10}(\log_3 10)) = 5$  then find the value of  $f(\log_{10}(\log_{10} 3))$ .
11. Let  $f(x)$  be a periodic function with period ' $p$ ' satisfying  
 $f(x) + f(x+3) + f(x+6) + \dots + f(x+42) = \text{constant } \forall x \in \mathbb{R}$ , then sum of digits of ' $p$ ' is
12. Find the inverse of  $f(x) = 2^{\log_{10} x} + 8$  and hence solve the equation  $f(x) = f^{-1}(x)$ .

MFN227

MFN228

MFN229

### ANSWER KEY

| Question | 1  | 2  | 3 | 4 | 5 | 6 | 7 | 8 | 9                                                                    | 10 |
|----------|----|----|---|---|---|---|---|---|----------------------------------------------------------------------|----|
| Answer   | B  | A  | D | C | B | C | B | B | $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 2$ | 3  |
| Question | 11 | 12 |   |   |   |   |   |   |                                                                      |    |
| Answer   | 9  | 10 |   |   |   |   |   |   |                                                                      |    |



Functions : RACE-08

Single Correct Question

1. The fundamental periods (if periodic) of the function,  $f(x) = \ln(\cos x) + \tan^3 x$  is  
 (A)  $\pi$  (B)  $2\pi$  (C)  $3\pi$  (D)  $4\pi$  MFN230
2. The fundamental periods (if periodic) of the function,  $f(x) = e^{x-[x]}$ ,  $[.]$  denotes greatest integer function is  
 (A) 1 (B) 2 (C) 3 (D) 4 MFN231
3. The fundamental periods (if periodic) of the function,  $f(x) = \left| \sin \frac{x}{2} \right| + \left| \cos \frac{x}{2} \right|$   
 (A)  $\pi$  (B)  $2\pi$  (C)  $3\pi$  (D)  $4\pi$  MFN232

Matrix Match Type Question

4. Match the function mentioned in column-I with the respective classification given in column-II. (where  $[.]$  and  $\{.\}$  denote greatest integer function and fractional part function respectively)

Column-I

Column-II

- |                                                                            |               |
|----------------------------------------------------------------------------|---------------|
| (P) $f: R \rightarrow R^+ f(x) = (e^{[x]})(e^{\{x\}})$                     | (1) one-one   |
| (Q) $f: (-\infty, -2) \cup (0, \infty) \rightarrow R f(x) = \ln(x^2 + 2x)$ | (2) many-one  |
| (R) $f: [-2, 2] \rightarrow [-1, 1] f(x) = \sin x$                         | (3) onto      |
| (S) $f: R \rightarrow R f(x) = x^3 - 3x^2 + 3x - 7$                        | (4) periodic  |
|                                                                            | (5) aperiodic |

MFN233

Integer Type Question

5. Suppose that  $f$  is an even, periodic function with period 2, and that  $f(x) = x$  for all  $x$  in interval  $[0, 1]$ . Find the value of  $f(3.14)$ .

MFN234

ANSWER KEY

| Question | 1 | 2 | 3 | 4                                  | 5    |
|----------|---|---|---|------------------------------------|------|
| Answer   | B | A | A | P→1,3,5; Q→2,3,5; R→2,3,5; S→1,3,5 | 0.86 |